

Zero Copy Cloning



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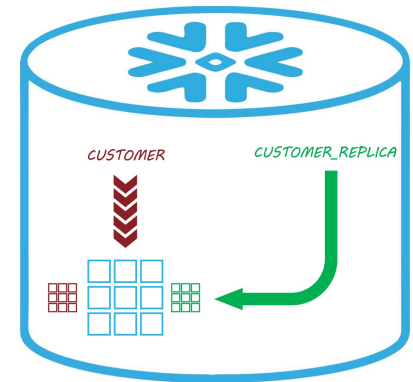
Cloud Data Architect
Snowflake Snowpro Certified

Section Introduction-Zero Copy Cloning

- What is Zero Copy Cloning ?
- What are the advantages of Zero Copy Cloning ?
- Zero Copy Cloning use cases.
- Privileges Inheritance in Zero Copy Cloning.
- LAB 1-Zero Copy Cloning of Tables.
- LAB 2-Zero Copy Cloning of Sequence, File Formats, Stages, Streams and Tasks.
- LAB 3-Zero Copy Cloning of Database and Schema.
- LAB 4-Zero Copy Cloning with Time Travel.
- Summary.

What is Zero Copy Cloning?

Zero Copy Cloning is a Snowflake feature that allows Data Engineers to create a copy of a database schema or table instantly without duplicating the underlying data.



What are the Advantages of Zero Copy Cloning?

Instant Creation & Time Saving

Cloning is instantaneous and is a metadata operation because it does not involve copying data. The Clone references the same micro-partitions as the original object. No actual copying of data happens. This leads to saving Data Engineer time.

No Additional Storage costs Initially

Since the clone shares the same data as the original or source object, it does not consume additional storage at the time of creation. Additional storage is needed only when data changes in either the original table or the clone.

Credit Saving

Since Zero Copy cloning is a metadata operation, there is no credit consumption. This leads to tremendous saving of cost as compared to traditional databases.



Zero Copy with Time Travel

Zero Copy Cloning supports time travel. You can create a database, schema or table with the state of the data in the past.

Zero Copy Cloning Use Cases

Refreshing Lower Environments

Lower environments like DEV and QA need regular refreshing with data from PROD. Zero Copy cloning can be used to create new DEV and QA environments(Database) after which the old DEV/QA environment can be dropped.

Data needed beyond Time Travel Expiration.

The maximum time travel retention period is 90 days, if you need to store a copy of a database, schema or table beyond 90 days. In that case you can use Zero Copy cloning to create a never expiring copy of the data.

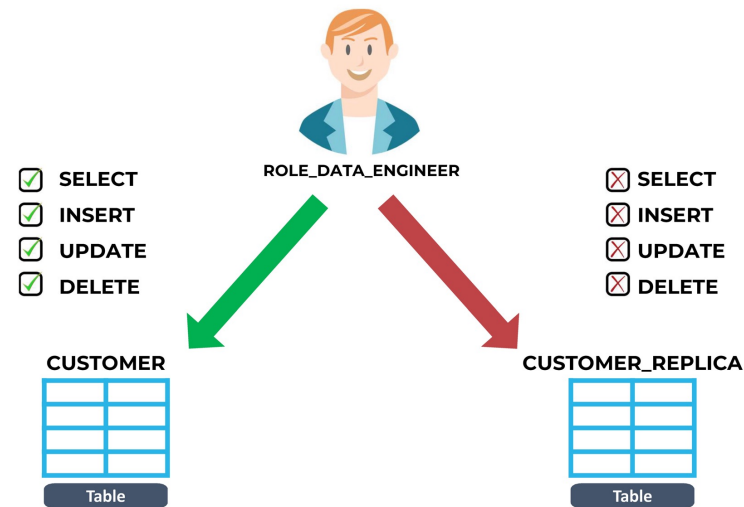
Need for static data.

Execution of Artificial Intelligence(AI)/Machine Learning(ML) generally needs static data. If AI/ML is run on changing data it will lead to model instability and unreliable predictions.



Privileges Inheritance in Zero Copy Cloning

- When cloning a table the access privileges granted on the source table is not inherited by the cloned table.
- When cloning a database or schema all child objects contained in the cloned database or schema inherits all the privileges granted on the child objects contained in the source database or schema.
- The database or schema does not inherit the privileges granted on the source container.



Summary

- Welcome to Section summary for ZCC.
- Zero Copy Cloning is a Snowflake feature that allows Data Engineers to create a copy of a database schema or table instantly without duplicating the underlying data.
- Databases , Schema, Tables, Sequences, File Formats, Streams and Tasks can be cloned in Snowflake.
- When a table is cloned in Snowflake the table initially shares the storage with the source table and hence does not consume any storage of its own.
- Zero Copy Cloning is a metadata operation and hence cloned object creation is instant and does not consume credits.
- Zero Copy supports Time Travel and can be used to restore data back to a point in the past.
- The most popular use cases for Zero Copy Cloning is refreshing DEV/QA databases with data from production, saving data beyond max time travel retention period of 90 days and using static data for AI/ML.
- When cloning a table the access privileges granted on the source table is not inherited by the cloned table.
- When cloning a database or schema all child objects contained in the cloned database or schema inherits all the privileges granted on the child objects contained in the source database or schema.
- The database or schema does not inherit the privileges granted on the source container.

Summary

- A Transient table cannot be cloned to a Permanent table.
- A Temporary table cannot be cloned to a Permanent table.
- A Permanent table can be cloned to a Permanent table.
- A Permanent table can be cloned to a Transient table.
- A Permanent table can be cloned to a Temporary table.
- A Temporary table can be cloned to a Temporary table.
- A Temporary table can be cloned to a Transient table.
- A Transient table can be cloned to a Transient table.
- A Transient table can be cloned to a Temporary table.
- When a table which uses a sequence to populate one of its columns is cloned the sequence is not cloned with the table.
- When a stage with a File Format is cloned, the cloned stage uses the same file format.
- When a stream is cloned all the rows in the stream are also cloned.
- Streams are cloned so that Stream data can be used in more than one ETL branch.
- Tasks are cloned to version Tasks so that they can be referred to if there is an issue with the ETL load related to scheduling.
- When a schema is cloned all the objects in the schema are also cloned. This happens even if it contains objects like Snowpipe, views or Stored Procedures which are not cloneable at the individual object level.
- When a database is cloned all the schemas inside the database are cloned and all objects inside the cloned schema are cloned too.

Thank you



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